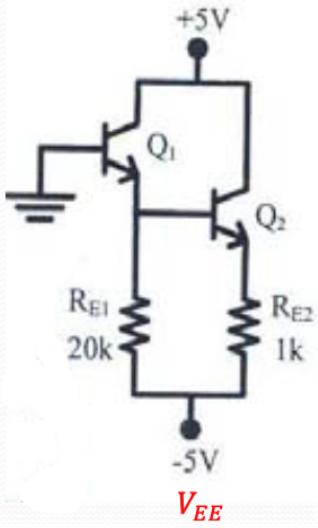
Implement the given logical functions using minimum CMOS logic. Perform transistor sizing such that the performance of the circuit matches the basic CMOS inverter with transistor size n and p.
Assume that the input and their complements are available.

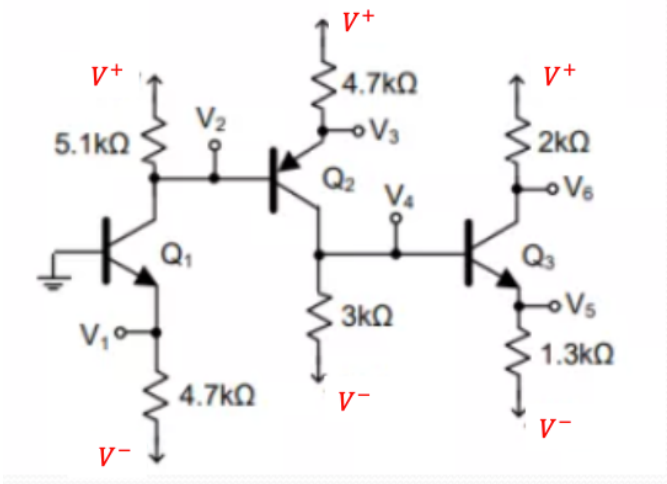
$$F_1(A,B,C,D) = \overline{ABD} + \bar{B}(\overline{A+CD})$$

$$F_2(x,y,w,z) = \overline{(x+w+z)}\,.\,w(\overline{y+xz})$$

Analyze the DC circuit below. Find all circuit variables. Do not ignore the base current in any of the two transistors. Both transistors have $\beta = 80$.

Keeping everything else the same, what should be the range of values of V_{EE} to saturate both transistors?
Show your work.

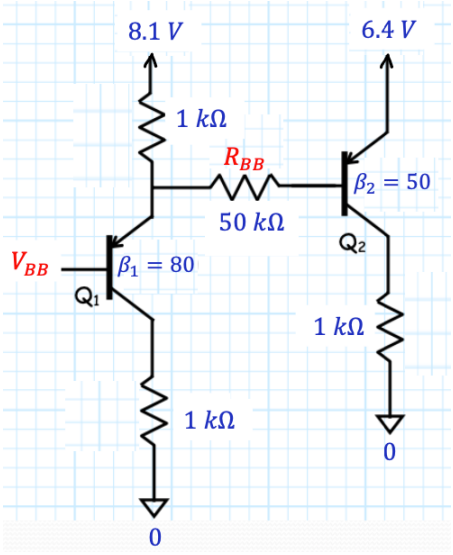




Find the voltages V_1 to V_6.
Do not ignore the base current in any of the transistors.

Show all your work.

Take V^+=15 V and V^{---}=-10 V.

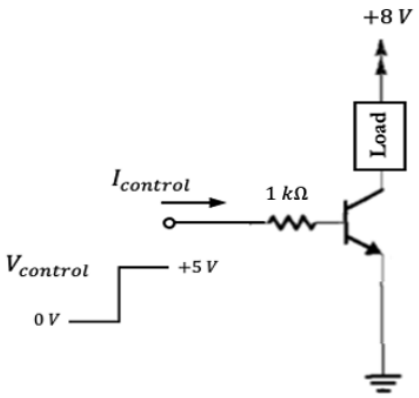


Analyze the DC circuit below. Find all circuit variables. Do not ignore the base current in any of the two transistors.

Keeping everything else the same, what should be the range of values of R_{BB} to saturate both transistors? Show your work.

V_{BB}=5 V

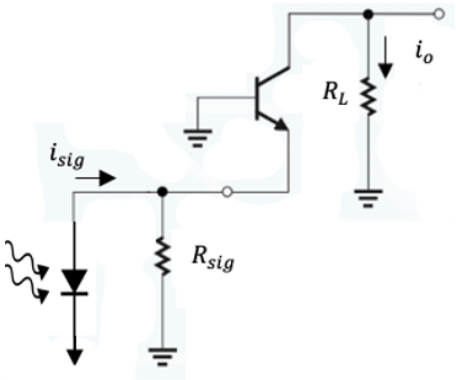
In the given circuit, the BJT has a base-emitter ON voltage of 0.7 V, a base-collector ON voltage of 0.5 V and β=50. Assume that the load is a lamp. The circuit is supposed to switch the lamp ON or OFF based on a digital control signal ($V_{control}$, see figure). Answer the following questions with full details.



a) The maximum possible voltage <u>across the lamp</u> is (circle one)
(a) +8 V (b) +7.8 V (c) +7.5 V (d) +5 V (e) +4.3 V.
Explain:

b) What are the values of $V_{control}$ and $I_{control}$ to achieve the voltage in part (a)?
c) Show that the minimum lamp resistance to achieve the voltage in part (a) is about 36 Ω.
d) If the lamp resistance is 50 Ω, what is the <u>maximum</u> power consumed by the lamp?
e) If the lamp resistance is 15 Ω, what is the <u>maximum</u> power consumed by the lamp?

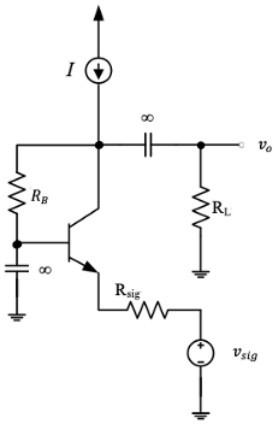
The input to the amplifier circuit below is a weak photodiode current i_{sig} (considered as an independent ac current source). The photodiode has an internal resistance R_{sig}, as shown. Assume that the transistor is in active mode, ignore r_o and,



a) Draw the amplifier equivalent circuit using the <u>T-model</u> for the BJT.
b) Show that the amplifier current gain $A_i = \alpha$.
c) Find the overall amplifier current gain.
d) Find the amplifier input resistance.
e) Find the amplifier output resistance.
f) Show that the amplifier voltage gain $A_v = g_m R_L$.
g) Based on your results, does this circuit serve as a "good" current amplifier for this application? Explain.

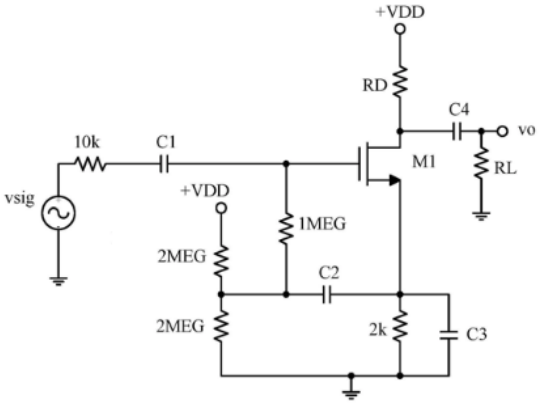
For the amplifier circuit below, with the transistor in active mode, ignore r_o and,

a) Draw the amplifier equivalent circuit using the pi-model.
b) Find the amplifier voltage gain.
c) Find the overall amplifier voltage gain.
d) Find the amplifier input resistance.
e) Find the amplifier output resistance.
f) Find the amplifier current gain.
g) What is the amplifier type? Do your results confirm the amplifier type? Explain.



Consider the MOSFET amplifier circuit below. Assume that the transistor is in saturation mode, ignore r_o and,

a) Draw the small-signal circuit of the amplifier using the MOSFET pi-model and the T-model.
b) Derive an expression for the voltage gain.
c) Derive an expression for the overall voltage gain.
d) Derive an expression for the current gain.
e) Find the amplifier input resistance.
f) Find the amplifier output resistance.
g) Based on your results, does this circuit serve as a "good" voltage amplifier or as a "good" current amplifier? Explain.



Consider the amplifier circuit below. Both transistors are saturated (no need for DC analysis). For both transistors λ=0. Take capacitors to be large ($C \to \infty$).

1) Draw the amplifier equivalent circuit using the pi-model for both transistors.
2) Find the amplifier voltage gain.
3) Find the amplifier input resistance.
4) Find the amplifier output resistance.

